

Ryan Li

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EDUCATION

University of Waterloo

Waterloo, ON

Computer Science and Combinatorics & Optimization (CO) Double Major

- Wat Street, WAT.ai, Data Science Club, Quant Club, Poker Club **GPA: 96.0% (97.0% Faculty Average)**

TECHNICAL SKILLS

Languages: Python, C/C++, Java, R, SQL, TypeScript, Rust

ML/Data: PyTorch, scikit-learn, Hugging Face, NumPy, SciPy, pandas, OpenCV, Pillow, W&B, Modal

Web/Infra/Tools: Flask, FastAPI, Express.js, React, Git, Docker, Kubernetes, PostgreSQL, Vercel, AWS

HONORS & AWARDS

- **National Mathematics Scholarship (\$15,000)**, President's Scholarship of Distinction (\$2,000)
- **Governor General's Medal** (graduated 1/400+), IB Diploma 44/45 (**top ~0.5% worldwide**)
- **Canadian Mathematics Olympiad Qualifier, Fermat Winner (placed 1/19644)**, 3× AIME, Putnam Top 25%

EXPERIENCE

Environment and Climate Change Canada

May 2026 – Present

Software Engineer Intern (Hydrometric Data Systems)

- **Spearheaded the Work Planning Tool** used by **300+ field technicians** across Canada; engineered ETL pipelines to sync Aquarius hydrometric data into a Flask/MariaDB backend with AWS S3-backed storage.
- Designed a DAG longest-path algorithm (topological sort + dynamic programming) to reconcile overlapping Aquarius approval entries, maintaining consistent records over **2,800+ monitoring stations**.
- Architected an **agentic harness** (LangGraph, AWS Bedrock) that dynamically routes between query-planning and tool-execution nodes via conditional edges to automate data queries for technicians.
- Introduced Docker layer caching into the existing CI/CD pipeline, **cutting deployment time by 85%**.

WAT.ai (University of Waterloo)

May 2026 – Present

Machine Learning Researcher

- Implemented **SPOC pseudo-labeling** for object state-change segmentation, **closing 55% of the gap** between a no-pseudo-label baseline and an mIoU oracle ceiling of 0.503. Ran inference on Modal serverless GPUs, tracking experimentation with Weights & Biases for reproducible team-wide results..

Wat Street (University of Waterloo)

Dec 2025 – Apr 2026

Quantitative Developer Intern

- Built a Python pipeline (yfinance → pandas → Parquet) detecting flagpole impulses via **volatility-scaled z-scores** (**≥1.8**) across 5-min bars; grid-searched the flagpole window size (4 bars) for sample adequacy and robustness.
- Trained gradient boosting classifiers using purged walk-forward CV and Spearman/PFI/SHAP-selected features to predict trend continuation, **filtering 39% of low-quality flags** versus a hold-based baseline.

PROJECTS

AlphaHex - Catan Engine | Python, PyTorch, NumPy, FastAPI, Monte Carlo Tree Search

- Trained a 1v1 Catan engine with a self-play MCTS-NN pipeline to learn board-state evaluation and expansion strategy, achieving a **top 500 colonist.io rank** with a 70% win rate.

Slotify (UofTHacks Winner) | Python, TypeScript, Node.js, React, FFmpeg, ElevenLabs

- Built a B2B web app and API for natural insertion of sponsor reads into podcasts and audiobooks, integrating OpenAI API for contextual ad generation and ElevenLabs API for voice cloning; **reduced processing latency by 35%** by supplementing LLM-based loudness and rhythm normalization with FFmpeg/librosa/pydub pipelines.

Provenance (cuHacking Winner) | TypeScript, Solana/Anchor (Rust), MongoDB Atlas, Chrome Extension (MV3)

- Developed an image authenticity system that cryptographically signs photos at capture and anchors them on-chain (Solana), achieving **sub-second (~400ms) trustless verification** in a single RPC.
- Extended verification beyond hash-matching to surfacing visually similar reposts via MongoDB Atlas Vector Search over CLIP embeddings.